Mechanical sensors

Graphene-Based Smart Insole Sensor for Pedobarometry and Gait Analysis

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Abstract—Pedobarography and gait analysis are crucial components of healthcare, given their close association with chronic illnesses. Wearable electronics, such as smart insole systems, have emerged as a promising technology for remote and long-term gait monitoring, providing more accurate data in real time. To meet the demands of wearable electronics, components with both flexibility and sensitivity are crucial. This letter employed low-cost and rapid prototyping techniques to develop a flexible piezoresistive pressure sensor with a graphene nanoplatelet/ polycaprolactone composite spray-deposited on a fiber-based commercial insole. The sensor demonstrated a pressure range of up to 400 kPa and high linear sensitivity (∼0.376 kPa-1), making it suitable for pedobarography. In addition, to test its practical application, the sensor was linked to a low-power programmable data logging and transmitting device for data acquisition over custom designed android application. The sensor can differentiate various motion states, such as walking, jumping, and running, while also demonstrating potential for long-term stability.

Index Terms—Mechanical sensors, gait monitoring, graphene, pressure sensor, smart insole, wearable technology.

I. INTRODUCTION

Recently, wearable electronic systems have been rapidly evolving in the areas of human motion detection, sports monitoring, and gait diseases as they are a promising replacement for traditional devices. Smart insoles can clarify human gait, and plantar pressure variations, and provide real-time evaluation of sports-related health, all contributing to clinical diagnosis and the reduction or elimination of sports-related injuries [1], [2], [3], [4]. To this aim, various wearable systems have been developed for long-term and long-distance analysis such as inertial measurement units (IMU) [5], [6]. However, these systems have a level of obstacles that make them insufficient to be used in this field. For instance, the accuracy of measurement in the IMU system depends on developing complicated models and algorithms.

In particular, the integration of pressure sensors has increased attention to realizing smart insoles capable of monitoring plantar pressure and/or gait cycle. Tao et al. [7] designed a smart insole with the insertion of a pressure sensor in an elastic dielectric layer (Ecoflex film) through laser-cut vertical pores. Its all-in-one system is capable of real-time monitoring of plantar pressure and differentiating between various standing postures and yoga poses. Guo et al. [8] developed an insole using carbon black/silicone rubber composite acting as a capacitive pressure sensor to detect plantar pressure. Although these developed smart insoles have been effective in human gait assessment, most of them exhibit poor functionality or produce unstable signals, and require complicated fabrication procedures. Integrating sensors fully into the insole is typically challenging [9], [10], resulting in high costs, discomfort, and overall dissatisfaction for users. These problems hinder the further progress and implementation of the device.

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To achieve an effective monitoring system, the investigation of sensor technology principles, and insole structure, and then addressing the challenges of easily and rapidly integrating a wearable sensor into the insole, is crucial [11], [12].

Generally, the piezoresistive mechanism has gained popularity due to its simplicity, information conversion, and electromechanical capabilities, whereas it faces some limitations. For instance having a broad pressure range along with high sensitivity is challenging. Moreover, the fabrication process of some developed sensors is complicated and expensive [13]. Hence, the development of a successful smart pressure-sensing insole for a wearable device requires overcoming these problems while maintaining flexibility and stretchability, an easy and cost-effective fabrication method, a constant signal, and a high level of accuracy.

With the aim of overcoming the challenges posed by existing sensors, a graphene nanoplatelets (GNPs)/polycaprolactone (PCL) composite is used for the realization of a highly flexible piezoresistive pressure sensor on the surface of a commercial fiber-based insole, acting as the substrate. GNPs are graphene-based nanofillers composed of stacked graphene layers with a thickness below 10 nanometers and lateral dimensions in the micrometer range [14]. They are chosen for their cost-effective nature, ease of fabrication, and noticeable mechanical, thermal, and electrical properties, while PCL is selected due to its biocompatibility and biodegradability [15].

The prototype sensor has the benefit of being easily and inexpensively produced, and is able to detect user movements over a broad range of pressure with reliable signals. Moreover, since it is produced through direct deposition of a thin graphene-based layer over a commercial insole, it does not compromise the user comfort and can be easily scaled up meeting industrial requirements. In addition, the produced graphene/based layers are known for their superhydrophobic properties [14].

The resulting GNP/PCL pressure sensor has a good linearity response up to 0.376 kPa^{-1} and a large pressure range of 0–400 kPa,

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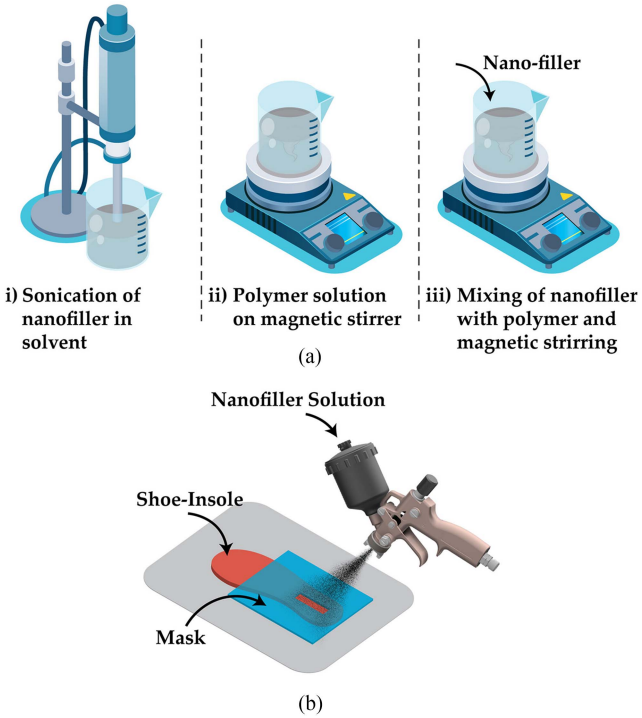

Fig. 1. (a) Fabrication process of GNP/PCL composite. (b) Spray coating of solution onto commercial fabric-based insoles.

enabling its applications in daily life. Moreover, this letter demonstrates the practical use of the designed pressure sensor, integrated into a shoe insole with a programmable wearable device, for detecting the movement of the user's foot. In particular, the sensor system is able to accurately identify motion states for activities such as walking, jumping, and running.

II. MATERIAL AND METHOD

A. Design and Fabrication of Pressure Sensor

The process of GNP/PCL composite fabrication is depicted in a schematic representation of Fig. 1(a). The GNPs are synthesized through the thermal expansion method in a furnace, starting from commercially available graphite intercalation compounds, and then dispersed in acetone using ultrasonication [14], [16]. The PCL powder is added to the acetone, and then stirred magnetically at 35 °C until the polymer dissolves. Finally, GNP solution is added to polymer, and the mixture is stirred for 30 min. Three different weight percentages of GNPs, 0.2%, 0.5% w/w, and 0.7% w/w are incorporated into the polymer matrix to produce Sensor-1, Sensor-2, and Sensor-3. Finally, the PCL/GNP mixture is sprayed onto fabric-based insoles, as shown in Fig. 1(b), and left to dry in the ambient environment.

B. Morphological and Electrical Characterization

A comprehensive morphological analysis of the membranes was carried out using Field Emission Scanning Electron Microscopy. A high-resolution sputter coater applied a 20 nm chromium layer to the samples before analysis.

Three rectangular sensors (2 cm × 6 cm) with different GNP contents (Sensor-1: 0.2% w/w, Sensor-2: 0.5% w/w, and Sensor-3: 0.7% w/w) are positioned beneath the heel area. The shorter edges of the sensors are coated with a thin layer of silver paint, and the wires are connected using a silver-based epoxy adhesive. The Keithley 2400 Source Meter dc source meter is used to obtain the electrical signals from the pressure sensors, while the universal testing machine INSTRON 3366 is used to measure the pressure forces.

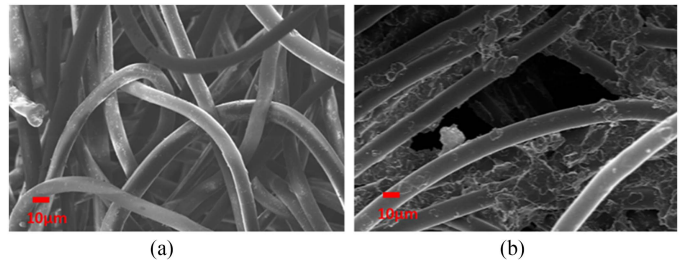

Fig. 2. SEM images of the commercial Insole before (a) and after (b) coating.

TABLE 1. Sensitivity of the Fabricated Pressure Sensors

PCL/GNP sensors	Highest Sensitivity	Pressure range
Sensor 1	*0.188 kPa-1*	*0 – 400 kPa*
Sensor II	*0.376 kPa-1*	*0 – 400 kPa*
Sensor III	*0.263 kPa-1*	*0 – 400 kPa*

The sensitivity of sensors is defined as

$$S = \frac{\delta\,(\Delta R/R_0)}{\delta P} \tag{1}$$

where R_0 is the initial resistance, and R is the measured resistance for the applied pressure P.

III. RESULTS AND DISCUSSION

A. Surface Morphology

The SEM surface images of the graphene/PCL nanocomposite coating that was prepared and applied over a commercial insole fabric are depicted in Fig. 2(a) and (b), respectively, before and after coating deposition. It is evident that the graphene nanocomposite layer covers the surface of the fiber in a way that is uniform. In addition to this, a successful integration of the GNPs into the polymer matrix has been observed.

B. Evaluation of Sensor Performance

In Fig. 3(a), the percent relative resistance variation $\Delta R/R_0$ is shown in response to the applied pressure P for Sensor-1, Sensor-2, and Sensor-3. Although all sensors display remarkable linearity in their electrical response, their sensitivity varies. Sensor-2 shows the highest sensitivity due to its GNP concentration, while sensors with lower and higher GNP content compromise sensitivity. In fact, Sensor-3 employs a greater amount of GNPs, leading to aggregation phenomena that compromise the signal quality. Despite the developed sensor being focused mainly on plantar pressure detections, the sensor is capable of detecting low-pressure changes with a detection limit of ≈1.5 kPa and a resolution of less than 1 kPa.

The main sensing properties of the pressure sensors are summarized in Table 1.

The fabricated sensors are tested for stability by subjecting them to multiple pressure loading and unloading cycles. Despite their impressive stability, the performance of the best sensor (Senosr-2) is reported. Fig. 3(b) shows the percent relative resistance variation $\Delta R/R_0$ (blue lines) in response to the applied pressure P load (red

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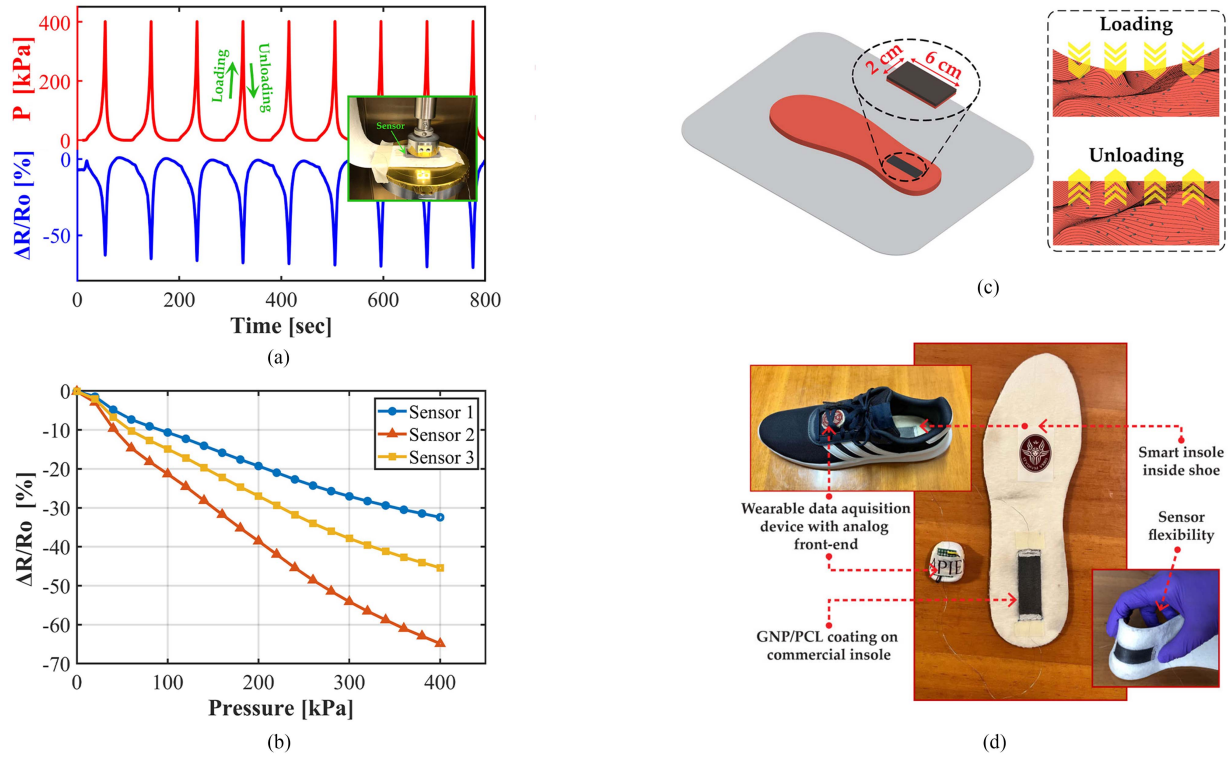

Fig. 3. (a) Percent relative resistance change ΔR/R0 under applied pressure P for Sensor-1, Sensor-2, and Sensor-3. (b) Variations in percent relative resistance change ΔR/R0 (blue lines) as a result of the applied pressure P load (red curve) over time for Sensor-2. (c) Schematic illustration of the composite on fabric-based insole under pressure. (d) Flexible insole sensor for real-time foot tracking.

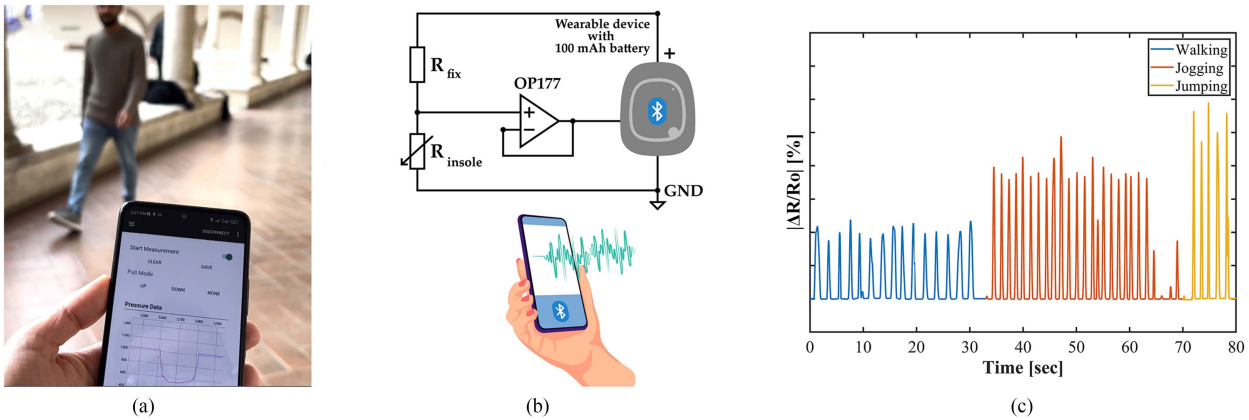

Fig. 4. (a) Smart insole system equipped with a personalized output signal acquisition mechanism and android application. (b) Recognition of and interpretation of the foot movements of the user. (c) Smart insole system for walking, running, and jumping foot state identification.

curve) over time. Sensor-2 exhibits excellent reproducibility, with almost identical percent relative resistance variation over eight cycles. Any small variations are attributed to GNPs rearrangement within the polymer matrix.

C. Working Principle

Fig. 3(c) shows a fabric-based insole with a pressure sensor. The sensor's performance depends on the concentration of conductive fillers and their microstructure [16], [17], [18], [19]. Under pressure, the GNPs in the fibers cluster together, increasing the number of direct contacts, leading to conduction paths and, as a result, higher electrical conductivity. However, the percolation threshold theory suggests a limit where conductivity stabilizes, leading to optimal piezoresistive behavior [20]. The sensor with 0.5% w/w of GNPs is found to perform best near this threshold. Higher concentrations can cause aggregation and compromise the sensor's response.

D. Insole Sensor Wearable Application

The effectiveness of the developed smart pressure sensor integrated inside a shoe insole [as shown in Fig. 3(d)] is evaluated by means of a wearable application test. The test is designed to capture the user's foot movements in real-time and transmit the data wirelessly to an Android application for further analysis.

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To measure the sensor's resistance change under load, a voltage divider circuit is utilized, which is comprised of a fixed resistor R_{fix} of value comparable to the sensor's resistance R_{insole}. However, the high resistance of the coated insole sensor leads to considerable voltage drop across the analog-to-digital converter (ADC), resulting in reduced accuracy. In this regard, the voltage divider circuit is connected to the ADC, followed by a voltage buffer using OP177 to ensure impedance matching and maximum power transfer, as shown in Fig. 4(b). Here, we used a wearable real-time data streaming and logging device MetaMotionS from MBIENTLAB for the recording of our sensor's data. This device is composed of a Bluetooth Low Energy 2.4 GHz transceiver along with 9-axis IMU sensors that operate on a 3.7 V 100 mAh lithium-ion battery. In addition, the device allows us to interface sensors externally to a 10-bit ADC unit that ensures accurate digitization of voltage changes. Consequently, the wearable device wirelessly transmits raw sensor data, at up to 100 Hz, to an Android-based mobile application developed in-house.

During the test, the participant wear the shoe insole equipped with the sensor and data acquisition system [as shown in Fig. 4(a)] while performing a range of activities such as walking, running, and jumping. An example of the recorded data for 80 sec of walking, jogging, and jumping is sketched in Fig. 4(c), indicating a visible difference in signal intensity due to varying amounts of force affecting the sensor surface.

Overall, the wearable application test demonstrates the potential of developed graphene-based smart insole sensors for a range of applications, including sports and fitness, healthcare, and biomechanics research due to its good linearity and large pressure range.

IV. CONCLUSION

In this letter, we successfully developed a graphene-based smart insole sensor by spraying a PCL/GNP composite over a commercial insole. The resulting piezoresistive pressure sensor showed high linearity and sensitivity, making it suitable for gait recognition.

The produced sensors exhibited a high linearity and sensitivity, up to 0.376 kPa^{-1} for pressures ranging from 0 to 400 kPa.

The sensor is flexible, adjustable, and capable of detecting human movements such as walking, jumping, and running. The findings suggest that this sensor could be a promising solution for monitoring and diagnosing foot biomechanics ailments.

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